Contents

1 A local Old English sequence from allophony to weak-tail nasal loss

1.1 Introduction

This section continues the same local stretch of the Old English sound-change sequence. It begins with labial allophony and Sievers-law syncope, moves through palatalization, the narrow glide-loss bridge before umlaut, and the main umlautal chapter, then turns into the later sequence of j-cluster coalescence, nasal dissimilation, back mutation, and two short weak-tail notes.

The sequence is continuous without pretending to be one single historical law. Some chapters here carry substantial consonantal or vocalic developments; others remain brief because they preserve smaller linking steps whose historical range is narrower.

2 B allophony and Sievers-law syncope

2.1 Historical discussion

The two changes gathered here belong to different historical categories, and they sit next to each other in the sequence that leads into Old English palatalization. The first is the positional alternation of Germanic **b*. Hogg states the Old English distribution clearly: /b/ is a stop initially, after nasals, and in gemination, while the same segment is otherwise realized as a voiced bilabial fricative (Hogg 1992, 101–2). Ringe and Taylor support the broader West Germanic background by treating Proto-West-Germanic **b* as a segment whose stop and fricative values depend on position (Ringe and Taylor 2014, 121), and Luick's spelling evidence shows the same labial fricative pattern in Old English (Luick 1914--1940, 107).

Sievers' Law belongs to a different part of the historical discussion. It is a prosodic and morphological adjustment in heavy stems, not a distributional allophone of a stop consonant. Adamczyk treats the Old English reflexes of the law as real historical material in weak verbs and related formations (Adamczyk 2001, 61–72). Fulk gives the compact comparative summary through familiar forms such as *biddan* 'ask', *sellan* 'give', and *nerian* 'save' (Fulk 2018, 127, §6.15). The point of keeping the two changes together is therefore practical and chronological. The behavior of Germanic **b* still needs a brief place in the book, and Sievers-law syncope is the last narrow feeder before the palatalization sequence begins in earnest.

2.2 SC049. Distribution of **b* after vowels and liquids (`PGmcBAllophony`)

The first rule formalizes the stop-fricative alternation of Germanic **b*.

```
define PGmcBAllophony [
    {*b} -> {*β} || PGmcStarVocalic _,
    {*b} -> {*β} || [{*l} | {*r}] _
] .o. [
    {*β} -> {*b} || _ {*b}
];
```

In prose, the rule says that **b* becomes a fricative after vowels and liquids, while geminate **bb* keeps the stop value.

Historically, this is the sort of narrow distributional statement that the handbooks place within the consonant system and discuss only briefly on its own. Even so, it matters because later derivations assume that the alternation is already in place. The clearest tested consequence appears in *reġnboga* 'rainbow'. If the rule is moved before the earlier linking-vowel adjustment, the derivation yields *reġnfoga* 'rainbow' rather than expected OE *reġnboga* 'rainbow'. This gives the earlier boundary `SC037 < SC049`. No equally sharp later lexical breakpoint emerges within the tested sequence, so the rule has no explicit later boundary within the present sequence.

2.3 SC050. Sievers-law syncope (`SieversLawSyncope`)

The second rule removes the Sievers-law **i* before **j* after a consonant.

```
define SieversLawSyncope [
    {*i} -> 0 || [EnglishStarConsonant | EnglishPalatalConsonant] _ {*j}
];
```

In plain language, the rule contracts the heavier **-CijV-** sequence to **-CjV-**. That is why it belongs to the historical aftermath of Sievers' Law and stands apart from the earlier stop-fricative distribution.

Its place in the sequence is clearer than that of the allophony rule. If the change is delayed until after velar palatalization before front vowels (`OEVelarPalatalization`), the cluster behind *streċċan* 'stretch' is affected too late. With PGmc **strákkijaną* in the wrong order, the derivation yields *strecċan* 'stretch'. The expected Old English form is *streċċan* 'stretch'. That is a real chronological consequence. No equally precise earlier lexical breakpoint fixes how far back the syncope must stand, so the historical picture remains asymmetric. The rule is secure as an immediate feeder into the palatalization zone, even though its earlier limit is less sharply bounded. The later boundary is therefore `SC050 < SC052`.

3 Palatalization of **sk* to **sc*

3.1 Historical discussion

The palatalization of **sk* to Old English **sc* is one of the recognizable early cluster changes in the larger palatalization zone. Campbell distinguishes the cluster from plain velars when he remarks that **sk* is especially prone to palatalization and assibilation (Campbell 1959, 278, §440). Hogg gives the same change a clearer structural place by treating **sk* beside the palatalization of plain velars and before the later West Saxon diphthongal developments (Hogg 1992, 106–7, 111–12). Ringe and Taylor make the same sequence explicit when they distinguish the earlier palatalization of velars and **sk* from the later diphthongization after already palatal consonants (Ringe and Taylor 2014, 213–16, §§6.4.1, 6.5.1).

Luick is especially useful for the larger frame. He treats the cluster change as part of a broader early movement toward palatal articulation, while still allowing later vowel consequences to form a different chapter of the history (Luick 1914--1940, 157, §168). Fulk's summary is the most concise warning against overextension: Old English **sc* is palatal except in the well-known back-vowel environments that preserve harder outcomes (Fulk 2018, 28). The result is a historically clear rule, but not an excuse to merge the whole palatalization and umlaut region into one undivided chapter.

3.2 SC051. Palatalization of **sk* to **sc* (`OESkPalatalization`)

The implementation states the **sk* rule explicitly.

```
define OESkPalatalization [
    {*s} {*k} -> {*ʃ} || .#. _
] .o. [
    {*s} {*k} -> {*ʃ} || EnglishStarFrontVowel _ (EnglishStarConsonant | .#.)
] .o. [
    {*s} {*k} -> {*ʃ} || (EnglishStarConsonant | .#.) _ EnglishStarFrontVowel
] .o. [
    {*s} {*k} -> {*ʃ} || _ {*j}
] .o. [
    {*s} {*k} -> {*ʃ} || {*j} _
];
```

In prose, the rule turns **sk* into a palatal outcome in the environments that lead to Old English **sc*.

Its historical place is between the earlier restoration and the later palatal vowel developments. If it is moved too early, the forms behind *flasce* 'flask' and *wascan* 'wash' are fronted too soon, yielding *flæsce* 'flask' and *wæscan* 'wash' rather than expected OE *flasce* and *wascan*. This gives the earlier boundary `SC046 < SC051`. If it is moved too late, the cluster no longer feeds the later West-Saxon diphthongal outcomes that appear in *sċeaft* 'shaft', *sċēar* 'shear', *sċēaþ* 'sheath', *sċēap* 'sheep', and *sċield* 'shield'. That is why the rule sits naturally beside velar palatalization before front vowels (`OEVelarPalatalization`) and before West Saxon palatal diphthongization (`OEWsPalatalDiphthongization`).

No single later wrong form is isolated for the whole group of **sċea-** / **sċie-** witnesses, but the current notes do show that the cluster must already be palatalized before the later West-Saxon diphthongal rule applies. This gives the later boundary `SC051 < SC056`.

The narrower chapter shape matters. The cluster rule is real and historically visible, but it is still only one part of the broader palatalizing sequence. The change should therefore be read as a distinct cluster development inside that sequence, not as a complete account of Old English palatalization.

4 Velar palatalization before front vowels

4.1 Historical discussion

Luick places the change inside a broad early palatalizing movement. Under the heading “Frühe Verschiebungen in palataler Richtung,” he treats English `k` and `g` before bright vowels together with the larger field of palatal effects (Luick 1914--1940, 157, §168). His emphasis falls on the environment first: velars before bright vowels and in the vicinity of the palatal glide belong to one early phonological sequence. The examples associated with that sequence, such as *ceaster* ‘town’, *geaf* ‘gave’, *giefan* ‘give’, and *giest* ‘guest’, already show that consonantal palatalization and later vowel effects stand close together historically, even when they must be distinguished analytically (Luick 1914--1940, 157–67, §§168–182).

Campbell narrows the picture by distinguishing plain velars from the especially palatal-prone `sk` cluster. His remark that “[sk] is more prone to palatalization and assibilation than [k]” is brief, but it makes clear that different members of the larger palatal field need not behave identically (Campbell 1959, 278, §440). Elsewhere in the same part of the grammar he uses forms such as *cild* ‘child’, *dæg* ‘day’, *giefan* ‘give’, and *giest* ‘guest’, which show how palatalized velars, palatal influence, and later umlautal outcomes meet in the same region of the lexicon without collapsing into one process (Campbell 1959, 69–72, 89, §§170, 190–191).

Hogg makes the conditioning sharper still. He states that the change takes place when the velar consonant is adjacent to and in the same syllable as a front vowel or the palatal consonant `j` (Hogg 1992, 103–4). This formulation is important because it moves the discussion from a broad list of palatal outcomes to a more precise phonological environment involving adjacency and syllable structure.

Ringe and Taylor make the chronological relation still clearer. When they write that “after initial velars and **sk* had been palatalized” West-Saxon diphthongization follows, plain velar palatalization becomes an earlier consonantal stage presupposed by later vowel developments (Ringe and Taylor 2014, 215, §6.5.1). Their own examples of the plain-velar rule, such as *weccan* ‘wake’, *licgan* ‘lie’, *lecgan* ‘lay’, *secg* ‘retainer’, *ecg* ‘edge’, *wicg* ‘horse’, and *brycg* ‘bridge’, illustrate the same point in lexical detail: front vowels and `j` create the palatal environment in which plain `k` and `g` cease to behave as plain velars (Ringe and Taylor 2014, 213–14, §6.4.1).

Taken together, these accounts show a gradual tightening of focus. Luick treats palatalization as a broad early movement. Campbell distinguishes more sharply between plain velars and the `sk` complex. Hogg specifies the adjacency and syllable conditions more directly. Ringe and Taylor then place the plain velar change in an explicit sequence that leads forward to later West-Saxon diphthongization. The literature therefore supports two claims at once: the change belongs to a larger palatalizing environment, and it must be kept distinct from neighboring processes if the sequence of developments is to be described accurately.

4.2 SC052. Palatalization of **k* before front vowels and **j* (`OEVelarPalatalizationKFro`

The first part of the implementation isolates the `k`-side environments of the change.

```
define OEVelarPalatalizationKFront [
    {*k} -> {*ʧ} || .#. _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || _ [{*i} | {*ī}],
    {*k} -> {*ʧ} || _ {*ī́},
    {*k} -> {*ʧ} || [{*i} | {*ī}] _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || {*ī́} _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || [{*i} | {*ī}] _ .#.,
    {*k} -> {*ʧ} || {*ī́} _ .#.
] .o. [
    {*k} {*k} -> {*ʧ} {*ʧ} || _ {*j}
] .o. [
    {*k} -> {*ʧ} || _ {*j}
] ;
```

In prose, the rule turns plain k into a palatal outcome before front vowels and j, including the geminated environment before j.

Historically, this section corresponds to the core of the older discussion of palatalized velars. It captures the environments behind forms such as *weccan* ‘wake’, *licgan* ‘lie’, and *lecgan* ‘lay’, where front vowels or j trigger the palatal outcome in the first place (Ringe and Taylor 2014, 213–14, §6.4.1). It is also the part of the process that prepares forms later assumed by velar palatalization before front vowels (OEVelarPalatalization) and, farther on, by fronting under i-umlaut (OEIUmlautFronting).

Within the present implementation, this helper rule is not ordered separately from the broader velar-palatalization rule below. Its chronology is therefore that of the larger rule it feeds. If the palatalization complex is moved before Sievers-law syncope, PGmc **strákkijaną* yields *streċċan* ‘stretch’ rather than expected OE *streċċan* ‘stretch’. If it is delayed beyond the umlautal core, PGmc **kūi* and **lúnganjō* yield *ċȳ* ‘cows’ and *lunġen* ‘lungs’ rather than expected OE *cȳ* and *lungen*. The shared boundary pattern is therefore SC050 < SC052 < SC055.

4.3 SC052. Velar palatalization before front vowels (OEVelarPalatalization)

The broader rule adds the g environments and composes them with the k palatalization rule above.

```
define OEVelarPalatalization [
    OEVelarPalatalizationKFront
] .o. [
    {*g} -> {*ʤ} || _ EnglishStarFrontVowel,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ .#.,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ EnglishStarFrontVowel,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ [EnglishStarConsonant - {*j}],
    {*g} {*g} -> {*ʤ} {*ʤ} || _ {*j}
] .o. [
    {*g} -> {*ʤ} || _ {*j}
];
```

In prose, the rule palatalizes plain k and g in front-vocalic and j-adjacent environments. Writing it as a separate rule clarifies the relative order of plain-velar palatalization, sk-palatalization, and umlautal developments.

The rule belongs after the earlier syncope that prepares forms like *streċċan* ‘stretch’ and before the later umlautal rules that would otherwise over-palatalize forms such as *cȳ* ‘cows’ and *lungen* ‘lungs’. See fronting under i-umlaut (OEIUmlautFronting) and the composite i-umlaut rule (OEIUmlaut) below.

If the rule is moved too early, before the syncope that prepares the consonant cluster, it breaks the derivation that should yield *streċċan* ‘stretch’. With PGmc **strákkijaną* in the wrong order, the model produces *streccan* ‘stretch’; the expected Old English form is *streċċan* ‘stretch’.

If it is moved too late, after i-umlaut, it over-palatalizes forms such as *cȳ* ‘cows’ and *lungen* ‘lungs’. PGmc **kūi* then yields *ċȳ* ‘cows’; the expected form is *cȳ* ‘cows’. PGmc **lúnganjō* yields *lunġen* ‘lungs’; the expected form is *lungen* ‘lungs’.

These lexical failures give the earlier boundary SC050 < SC052 and the later boundary SC052 < SC055.

Once the rule is in place, plain velars before front vowels and j no longer remain plain. They become the palatal outcomes presupposed by later developments, including the umlautal rules discussed in fronting under i-umlaut (OEIUmlautFronting). That matters for dictionary-like forms such as *cild* ‘child’ or *dæg* ‘day’ and for the broader relation between consonantal palatalization and later vowel-fronting processes (Luick 1914--1940, 157, §168; Campbell 1959, 278, §440; Ringe and Taylor 2014, 203–15, §§6.4.1, 6.5.1).

The evidence places the rule within a wider palatalizing environment, but it does not require every neighboring palatal process to be merged with it. sk belongs to a related but distinct development, and the later umlautal material poses a different historical problem. The relation to the earlier syncope rule is likewise specific and limited: the *strećċan* ‘stretch’ evidence shows a real dependency without turning the feeder process into a coequal sound law of the same scope.

5 The pre-umlaut bridge and loss of **w* before **i*

5.1 Historical discussion

The two rules gathered here are unequal in weight. The first is a narrow loss of **w* after velars in the **ngw* sequence. Ringe and Taylor make the historical core clear when they derive PGmc **singwan* to Old English *singan* 'sing' (Ringe and Taylor 2014, 214, §6.4.2). That gives the change a real comparative anchor, but it does not turn it into a large chapter of its own. It is the kind of small cleanup rule that needs a place in the sequence without claiming the status of a major handbook law.

The second rule is historically more legible. Campbell notes the recurring loss of **w* before **i* in unstressed position (Campbell 1959, 167, §406). Ringe and Taylor trace the development of *sǣ* 'sea' from earlier **saiwi-* / **sawi-* (Ringe and Taylor 2014, 257, §6.7.1), and Luick gives the same trajectory in his own historical grammar (Luick 1914--1940, 173, §187). The chapter therefore belongs in the stretch between plain palatalization and the umlautal core, but it should keep the asymmetry visible: the first rule is a narrow bridge, the second is a stronger glide-loss development with a specific lexical witness.

5.2 SC053. Loss of **w* after velars (`OEPostVelarWLoss`)

The first rule handles the **ngw* simplification.

```
define OEPostVelarWLoss [
    {*w} -> 0 || {*n} {*g} _
];
```

In prose, the rule removes **w* after the velar cluster in forms of the **singwan* type.

Historically, this is a very small rule. It keeps developments such as *singan* 'sing' visible in the sequence, but it does not create a large family of lexical breakpoints. Current testing does not recover a positive earlier or later boundary: the search reaches older material on the left and the later Old English search limit on the right with no decisive wrong form. If the rule is moved either earlier or later within the tested sequence, no lexical witness yet provides a sharper wrong/expected pair. The safest reading is therefore modest: this is a local bridge rule that belongs before the umlautal chapter without claiming a sharper chronological slot than the evidence supports.

5.3 SC054. Loss of **w* before final **i* (`OEWLossBeforeI`)

The second rule is the more historically legible member of the pair.

```
define OEWLossBeforeI [
    {*w} -> 0 || EnglishStarVocalic _ {*i} .#.
];
```

In prose, the rule removes non-initial **w* before final unstressed **i*.

The best witness is *sǣ* 'sea'. Campbell's discussion of the loss of **w* before **i*, Ringe and Taylor's derivation from earlier **saiwi-* / **sawi-*, and Luick's parallel account all point to the same historical consequence (Campbell 1959, 167, §406; Ringe and Taylor 2014, 257, §6.7.1; Luick 1914--1940, 173, §187). The glide has to disappear early enough for the preceding vowel to continue into the later fronted and lengthened outcome. If the glide survives too long, the derivation retains **w* and misses *sǣ* 'sea'. If the rule is moved before SC020, the same witness yields *sǣw* 'sea' rather than expected OE *sǣ*. This gives the earlier boundary `SC020 < SC054`. If the rule is delayed until after SC063, the same witness again yields *sǣw* rather than expected *sǣ*. This gives the later boundary `SC054 < SC063`.

This is why the chapter belongs immediately before the broader umlautal developments discussed in the composite i-umlaut rule (`OEIUmlaut`). The two rules together form a genuine bridge into that later vowel chapter, but only the second has a strong lexical and handbook footing of its own.

6 The Old English i-umlaut and West Saxon palatal diphthongization

6.1 Historical discussion

Luick gives the change its traditional scale:

> Der wichtigste Fall von palataler Beeinflussung … war die Veränderung der urenglischen Vokale durch i oder j der Folgesilbe.
>
> (Luick 1914--1940, 166–67, §182)

Campbell gives the most compact classical formulation in English when he writes that "the process known as i-umlaut or i-mutation operates on practically all the sounds which it could theoretically affect in OE" (Campbell 1959, 69, §190). He immediately defines the core conditioning environment as a following `i` or `j`, and he goes on to trace the consequences across much of the vowel system, including forms such as *giest* 'guest', *giefan* 'give', *hierde* 'shepherd', and *ieldra* 'older' (Campbell 1959, 69–72, §§190–197).

Hogg continues in the same vein: "we come now to a change which is almost as uncontroversial as it is important" (Hogg 1992, 112). His examples, such as *bryd* 'bride', *trymman* 'strengthen', *bedd* 'bed', *ciest* 'chest', and *wiersa* 'worse', likewise emphasize that the change is a broad redistribution of vowel quality across the Old English vowel system (Hogg 1992, 112–14).

The narrower palatal-diphthongal material is described differently. Ringe and Taylor treat West-Saxon diphthongization after initial palatals as a distinct process (Ringe and Taylor 2014, 215, §6.5.1), and Fulk is even more explicit about its chronological delicacy when he calls it "diphthongization by initial palatal consonants (which precedes front umlaut but not breaking)" (Fulk 2018, 74, §4.13). Ringe and Taylor's examples such as *gieldan* 'pay', *scield* 'shield', and *scieppan* 'create' show that this narrower process is triggered by already palatal consonants and leads to specifically West-Saxon diphthongal outputs (Ringe and Taylor 2014, 215–16, §6.5.1).

The sequence of discussion is fairly clear. Luick, Campbell, and Hogg all give i-umlaut primary importance. Ringe and Taylor and Fulk then help separate that major change from the narrower West-Saxon diphthongization that stands beside it. The literature therefore establishes a large, system-wide umlautal change and a narrower adjoining process affecting words after initial palatals. That distinction matters because the two processes act in different environments and produce different lexical consequences.

6.2 SC055. Fronting under i-umlaut (`OEIUmlautFronting`)

The first component of the implementation handles the broad fronting of vowels under the influence of following `i` or `j`.

```
define OEIUmlautFronting [
    {*a} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ā} -> {*ǣ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*e} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*o} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ō} -> {*ē} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*u} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ū} -> {*ȳ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*á} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*é} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ó} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ú} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

In prose, the rule fronts and raises the relevant simple vowels when a following `i` or `j` provides the trigger.

Historically, this is the most central part of the umlautal development described by Luick, Campbell, Hogg, Ringe and Taylor, and Fulk. Within the present implementation it stands after velar palatalization before front vowels (`OEVelarPalatalization`) and before the narrower West-Saxon palatal-diphthongization rule discussed below.

The handbooks describe the same conditioning environment in different ways but with the same phonological consequence: a following high front vocoid triggers the fronting of earlier back vowels. That is why forms such as *fylgan* 'follow', *gylden* 'golden', *wyrm* 'worm', and *giest* 'guest' can all be treated inside the same formal rule even though they belong to different lexical classes (Ringe and Taylor 2014, 222, §6.6.1; Campbell 1959, 69–72, §§190–191).

The same ordering logic that governs the umlaut complex governs this component. If the umlautal rule set is moved before SC052, PGmc **kūi* yields *ċȳ* 'cows' rather than expected OE *cȳ*, and **lúnganjō* yields *lunġen* 'lungs' rather than expected OE *lungen*. At the other edge, the later West-Saxon diphthongization must follow the umlautal rule set: if that later rule is moved too early, PGmc **géftiz* yields *ġieft* 'gift' rather than expected OE *ġift*, and **skáiθiz* yields *sċǣþ* 'sheath' rather than expected *sċēaþ*. This gives the shared boundary pattern `SC052 < SC055 < SC056`.

As a component rule, it shares the chronology of the composite i-umlaut rule (`OEIUmlaut`).

6.3 SC055. Raising under i-umlaut (`OEIUmlautRaising`)

The second component handles the raising of umlauted æ to e.

```
define OEIUmlautRaising [
    {*æ} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

In plain language, this rule takes the fronted low vowel created by the earlier fronting rule and raises it further where the same umlaut trigger still holds.

Historically, this belongs inside the same broad i-umlaut development. It is part of the same chronological development and shares the evidence base of the composite i-umlaut rule (`OEIUmlaut`).

Like the fronting component, this raising rule belongs inside `SC052 < SC055 < SC056`. If the umlaut complex is moved before SC052, **kūi* yields *ċȳ* instead of expected *cȳ* and **lúnganjō* yields *lunġen* instead of expected *lungen*. If the later West-Saxon diphthongization is moved too early, **géftiz* yields *ġieft* rather than expected *ġift*, and **skáiθiz* yields *sċǣþ* rather than expected *sċēaþ*.

This narrower subrule matters because the sources do not describe umlaut as simple fronting alone. Campbell explicitly notes that the low front vowel changes again before `m` and `n` in most dialects (Campbell 1959, 69, §190), and Hogg likewise treats short front vowels as part of the same assimilatory system (Hogg 1992, 112).

6.4 SC055. Diphthongal outcomes under i-umlaut (`OEIUmlautDiphthong`)

The third component handles the diphthongal outcomes that also undergo i-umlaut.

```
define OEIUmlautDiphthong [
    {*ea} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ēa} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*io} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*īo} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*eo} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ēo} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*éa} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*éo} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ío} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

In prose, the rule states that diphthongal inputs are subject to umlaut as well: the vowel change is not confined to simple vowels.

This matters historically because the handbooks describe i-umlaut as a system-wide assimilatory development. The rule therefore stands inside the same chronological bracket as fronting under i-umlaut (`OEIUmlautFronting`) and raising under i-umlaut (`OEIUmlautRaising`), even though its outputs are shaped differently.

The relevant examples are the recurring West-Saxon `ie` forms cited in the handbooks, including *giest* 'guest', *giefan* 'give', and *hierde* 'shepherd' in Campbell and *ciest* 'chest' in Hogg (Campbell 1959, 69–72, 78–80, §§190–191, 248–251; Hogg 1992, 112–14). The present formalization keeps those diphthongal outcomes visible as a distinct part of the general umlautal development and does not leave them implicit under the broad description of fronting.

Chronologically, this component also shares the same evidence as the umlaut complex as a whole. If the umlaut complex is moved before SC052, it over-palatalizes **kūi* and **lúnganjō*; too-early West-Saxon diphthongization yields *ġieft* and *sċǣþ* instead of expected *ġift* and *sċēaþ*. The rule therefore belongs within `SC052 < SC055 < SC056`.

6.5 SC055. The composite i-umlaut rule (`OEIUmlaut`)

The implementation also defines a composite rule that composes the three preceding parts.

```
define OEIUmlaut OEIUmlautFronting
    .o. OEIUmlautRaising
    .o. OEIUmlautDiphthong;
```

In prose, this says that the implementation treats the umlaut as a sequence of fronting, raising, and diphthongal adjustments composed in order.

Chronologically, the composite rule must follow velar palatalization before front vowels (`OEVelarPalatalization`). If it is moved too early, forms such as *cȳ* 'cows' and *lungen* 'lungs' become over-palatalized. PGmc **kūi* yields *ċȳ* 'cows'; the expected form is *cȳ* 'cows'. PGmc **lúnganjō* yields *lunġen* 'lungs'; the expected form is *lungen* 'lungs'.

The same local network gives the later boundary. If West-Saxon palatal diphthongization is moved too early, PGmc **géftiz* yields *ġieft* 'gift' rather than expected OE *ġift*, and **skáiθiz* yields *sċǣþ* 'sheath' rather than expected *sċēaþ*. The composite umlaut rule therefore occupies `SC052 < SC055 < SC056`.

Those failures show that the broad umlautal rule needs an earlier terminus post quem in the palatalization sequence, even though it remains the main vowel change within the present chapter.

The composite rule is important because the literature presents the umlaut as a single historical development even while the implementation decomposes it into formal parts. The composite definition is the point at which the separate fronting, raising, and diphthongal effects are treated as one chronological event in the Old English sequence.

6.6 SC056. West Saxon palatal diphthongization (`OEWsPalatalDiphthongization`)

The narrower West-Saxon rule is treated separately from the broader umlautal complex.

```
define OEWsPalatalDiphthongization [
    {*æ} -> {*ea} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*ǣ} -> {*ēa} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*e} -> {*ie} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*ē} -> {*īe} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*é} -> {*íe} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*ḗ} -> {*īe} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.]
```

```
];
```

In prose, this rule diphthongizes certain vowels after already palatal consonants in West Saxon. It therefore has a narrower dialectal and chronological scope than the broader umlaut rule.

The historical evidence for that narrower scope is concrete. Ringe and Taylor illustrate the rule with forms such as *gieldan* 'pay', *scield* 'shield', and *scieppan* 'create', where an already palatal consonant triggers the diphthongal outcome (Ringe and Taylor 2014, 215–16, §6.5.1). Hogg's *giefan* 'give' and *sceap* 'sheep' material belongs to the same phonological zone (Hogg 1992, 108–9), while Fulk distinguishes this palatal-consonant-triggered diphthongization from the broad front-mutation process (Fulk 2018, 74, §4.13).

Its place is later than the composite i-umlaut rule (`OEIUmlaut`). If this rule is moved too early, the later ordering is constrained by forms such as *ġift* 'gift' and *sċēaþ* 'sheath'. PGmc **géftiz* then yields *ġieft* 'gift'; the expected form is *ġift* 'gift'. PGmc **skáiθiz* yields *sċǣþ* 'sheath'; the expected form is *sċēaþ* 'sheath'.

This gives the earlier boundary `SC055 < SC056`. No comparably sharp later boundary is available.

No tested lexical item provides a comparably precise later terminus ante quem. The available evidence therefore establishes the rule's relation to the earlier umlautal process much more clearly than it fixes a later point by which it must already have applied.

The two rules should accordingly be kept distinct. The broad umlautal rule accounts for a system-wide assimilatory change; the West-Saxon rule accounts for a narrower palatal-consonant-conditioned diphthongization whose chronological and dialectal scope is more restricted.

7 J-cluster coalescence

7.1 Historical discussion

This chapter belongs to the later part of the palatalization and fronting region. Campbell, Ringe and Taylor, and Fulk all discuss the same neighborhood of palatalized and fronted outcomes that underlies forms such as *bīeġan* 'bend' and *sēċan* 'seek' (Campbell 1959, 89, 107–8, §§170, 248–251; Ringe and Taylor 2014, 213–51, §§6.4.1, 6.5.1, 6.6.1–6.6.4; Fulk 2018, 65, 75, §§4.7, 4.13). None of them turns this later cluster adjustment into a major independent headline. The historical interest lies in the fact that it remains a real part of the sequence even though the larger palatalization and umlaut chapters carry more of the explanatory weight.

That narrower scale matters. Earlier chapters have already established the plain velar and **sk* palatalizations, and the umlaut chapter has already handled the major vowel consequences. The present rule is a later coalescence inside that same neighborhood. It deserves explicit prose because the lexical outcomes are clear, not because it eclipses the larger processes around it.

7.2 SC057. Coalescence of velar + **j* clusters (`OEJClusterCoalescence`)

The implementation keeps the later cluster coalescence very small and explicit.

```
define OEJClusterCoalescence (
    [{*g} {*j} -> {*ʤ}]
    .o. [{*k} {*j} -> {*ʧ}]
);
```

In prose, the rule coalesces **gj* and **kj* into the palatal outcomes that later surface in forms such as *bīeġan* 'bend' and *sēċan* 'seek'.

Its earlier dependency is clearer than its later limit. If the rule is moved before velar palatalization before front vowels (`OEVelarPalatalization`), the developments behind *bīeġan* 'bend' and *sēċan* 'seek' are lost. Related forms such as *fylġan* 'follow', *heċġ* 'hedge', and *sengan* 'singe' fail in the same broader palatalization zone. PGmc `*báugijaną` yields *bēaġan* 'bend' rather than expected OE *bīeġan*, and PGmc `*sōkijaną` yields *sōċan* 'seek' rather than expected *sēċan*. This gives the earlier boundary `SC052 < SC057`. No comparably sharp later lexical breakpoint emerges within the remaining sequence, so the chronology remains short and one-sided.

That modest shape is historically appropriate. The rule is a real later member of the palatalization region, but it does not need to absorb the umlautal chapter behind it or the nasal-dissimilation chapter that follows it. The later coalescence remains visible in the sequence once the larger neighboring chapters are already in place.

8 Nasal dissimilation

8.1 Historical discussion

Luick preserves individual outcomes such as *enetre* ‘yearling’ (with the spelling *enitre* in his text) without isolating a separate law around them (Luick 1914--1940, 166). Campbell likewise reaches forms such as *heofon* ‘heaven’ in a discussion of suffixal variation and does not set them off in any special section on nasal dissimilation (Campbell 1959, 155). Hogg mentions *heofon* ‘heaven’ in the course of his account of back mutation, again without isolating a separate law (Hogg 1992, 112).

Fulk supplies the clearest general formulation: “In the cluster mn, the first consonant tends to lose its nasality by dissimilation, though the results are hardly regular” (Fulk 2018, 121, §6.11). Ringe and Taylor stay close to the lexical evidence and note that *enetre* ‘yearling’ reflects “loss of the second **n* by dissimilation” (Ringe and Taylor 2014, 282).

The discussion therefore develops from scattered lexical observations to a more explicit but still cautious generalization. Luick preserves the kind of form the rule is meant to capture. Campbell and Hogg show that related outcomes enter the handbooks, but only incidentally, as part of larger accounts of other changes. Fulk makes the recurrent `mn` tendency explicit, while Ringe and Taylor provide an exact lexical case in *enetre* ‘yearling’. What emerges is a limited but recurring dissimilatory pattern whose scope is far smaller than that of the major Old English vowel laws.

8.2 SC058. Nasal dissimilation in short-vowel environments (`OENasalDissimilation`)

The implementation formalizes the change as a narrow rule applying in short vowel environments before a following `n`.

```
define OENasalDissimilation [
    {*m} -> {*f} || EnglishStarShortVowel _ EnglishStarShortVowel {*n}
     ↪  [EnglishStarShortVowel | .#.]
];
```

In plain language, the rule turns medial `m` into `f` in a restricted short-vowel environment before a following syllable containing `n`.

Historically, the rule captures the limited type of dissimilation reflected in forms such as *heofon* ‘heaven’, *fæstenn* ‘fasting’, and *enetre* ‘yearling’. It is much narrower than the major vowel changes and is best understood as a recurring but partly lexicalized pattern.

The relation between the sources and the formalization is correspondingly close but not exact. Fulk formulates the tendency at the level of `mn` clusters and illustrates it with *heofon* ‘heaven’ and *fæstenn* ‘fasting’ (Fulk 2018, 121, §6.11). Ringe and Taylor show the same kind of development in *enetre* ‘yearling’ (Ringe and Taylor 2014, 282). Campbell’s “heofon is for older hefzen” and Hogg’s sequence **hefon* > *heofon* preserve outcomes of the same kind as those modeled here (Campbell 1959, 155; Hogg 1992, 112). The formal rule is therefore narrower than the total set of handbook remarks: it models one plausible recurrent environment and does not claim to exhaust every dissimilatory development involving nasals.

Chronologically, the available tests do not identify a sharper position within the Old English sequence. When the rule is moved earlier, no lexical breakpoint appears before the inherited West-Germanic material that precedes the tested Old English changes. When it is moved later, the tests likewise fail to identify a more precise later boundary within the remainder of the Old English sequence.

No comparable pair of lexical failures fixes a narrower slot here. The present evidence therefore gives neither a precise terminus post quem nor a precise terminus ante quem for the rule within the tested sequence. No exact wrong early or late output is currently available for this chapter.

Even so, the rule has real interpretative consequences. It provides a place in the implementation for outcomes of the *heofon* ‘heaven’, *fæstenn* ‘fasting’, and *enetre* ‘yearling’ type discussed in the literature (Fulk 2018, 121, §6.11; Ringe and Taylor 2014, 282; Campbell 1959, 155; Luick 1914--1940, 166; Hogg 1992, 112). Without an explicit rule, those outcomes would be left to diffuse analogy or to unexplained exception lists.

The evidence points to a narrow dissimilatory tendency, especially in `mn`-type clusters and a small group of lexical outcomes. There is no support for a regular change operating across a broad phonological field. The rule is secure enough to model, but the available tests leave its position within the Old English sequence underdetermined.

9 Back mutation

9.1 Historical discussion

Back mutation is the substantive center of this part of the sequence. Campbell treats it as a later Old English diphthongizing development before following back vowels, and his examples already show why forms such as *heofon* 'heaven' are historically legible outcomes in their own right (Campbell 1959, 86, §207). Hogg treats the same development as a later change with clear parallels to breaking (Hogg 1992, 112). Ringe and Taylor sharpen the picture by distinguishing West Saxon forms such as *giefan* 'give' and *wefan* 'weave' from non-West-Saxon forms such as *geofad* and *weofan* (Ringe and Taylor 2014, 319, §6.9.4). Fulk likewise treats back mutation as a distinct historical phenomenon with its own profile beside the earlier umlautal changes (Fulk 2018, 69, §4.8).

That makes back mutation different from the short notes that follow it. Back mutation belongs to the same local stretch of the sequence, but it carries more historical weight and clearer lexical consequences. Even so, its later relation lies beyond this immediate stretch of the sequence, and the later weak-tail region is best kept as a forward reference only.

9.2 SC059. Back mutation before labials and liquids (`OEBackMutation`)

The implementation keeps the change as one explicit rule.

```
define OEBackMutation [
    {*e} -> {*eo} || _ [EnglishStarLabial | EnglishStarLiquid] {*u},
    {*æ} -> {*ea} || _ [EnglishStarLabial | EnglishStarLiquid]
     ↪  EnglishBackMutationTrigger,
    {*é} -> {*éo} || _ [EnglishStarLabial | EnglishStarLiquid] {*u}
];
```

In prose, the rule backs and diphthongizes earlier front vowels before a following labial or liquid plus a back-vocalic trigger.

Its chronology is real on both sides, but not equally local. The earlier side is already fixed by the preceding vowel and weak-tail material. If the rule is moved too early, forms such as **gébaną* produce *ġeofan* 'give'; the expected form is *ġiefan* 'give'. **stélaną* likewise produces *steolan* 'steal'; the expected form is *stelan* 'steal'. The later side is different. If the rule is pushed too far to the right, **wébaną* yields *weofan* 'weave'; the expected form is *wefan* 'weave'. That later edge is real, but it points beyond the present stretch of the sequence into the later weak-tail reductions, so here it should remain only a forward reference.

These lexical failures give the earlier boundary `SC048 < SC059` and the later boundary `SC059 < SC078`.

This is why the change can serve as the center here without implying that the following weak-tail notes belong to the same historical law. The rule marks a real local seam, but the section after it immediately becomes narrower.

10 West Saxon palatal umlaut

10.1 Historical discussion

The evidence is narrow enough that the discussion can stay brief. Campbell and Ringe and Taylor both support the development behind forms such as *miht* 'might' and *niht* 'night', while Fulk's broader chronology makes clear that this material belongs beside the umlaut and palatal-vowel region as a subordinate note beside it (Campbell 1959, 107–8, §§248–251; Ringe and Taylor 2014, 215–51, §§6.5.1, 6.6.1–6.6.4; Fulk 2018, 65, 75, §§4.7, 4.13).

That is why the note belongs here after back mutation even though its clearest historical tie still reaches back to the earlier umlautal chapter. The phenomenon is real, yet its place in the sequence is one-sided. The evidence is clear enough to state and narrow enough to remain brief.

10.2 SC060. West Saxon palatal umlaut before **h*-clusters (`OEWsPalatalUmlaut`)

The implementation treats the West Saxon change as one explicit rule.

```
define OEWsPalatalUmlaut [
    {*eo} -> {*i} || _ OEHCluster .#.,
    {*io} -> {*i} || _ OEHCluster .#.,
    {*ie} -> {*i} || _ OEHCluster .#.,
    {*eo} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*io} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*ie} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*éo} -> {*i} || _ OEHCluster .#.,
    {*ío} -> {*i} || _ OEHCluster .#.,
    {*íe} -> {*i} || _ OEHCluster .#.,
    {*éo} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*ío} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*íe} -> {*i} || _ OEHCluster EnglishStarFrontVowel
];
```

In prose, the rule reduces short diphthongs to **i* before the relevant **h* clusters.

The crucial point is its earlier dependency. The rule must follow the composite i-umlaut rule (`OEIUmlaut`), because if it is moved too early the forms behind *miht* 'might' and *niht* 'night' remain at the overdeveloped stage *mieht* and *nieht* rather than expected OE *miht* and *niht*. No comparably sharp later lexical breakpoint emerges within the remainder of the section. The note therefore belongs here as a short afterpiece to the umlaut chapter, not as the start of a new larger unit.

This gives the earlier boundary `SC055 < SC060`. No comparably sharp later boundary is available.

11 Weak-tail nasal loss

11.1 Historical discussion

The development belongs to the narrower end of the later weak-tail sequence. It is historically legible through the pathway that leads to *dōn* 'do', and the broader late weak-tail setting is supported by the usual handbook discussions of apocope and related reduction (Campbell 1959, 144–45, §§345–349; Hogg 1992, 120–21; Fulk 2018, 91, §5.6). But the decisive lexical tie lies much farther back in the sequence, in the older development of **dōną*. That keeps the note real, while also keeping it small.

Within this later run of changes it follows back mutation and West Saxon palatal umlaut, but the evidence remains slighter than theirs.

11.2 SC061. Reduction of final nasal weak-tail endings (`OEWeakTailNasalLoss`)

The implementation keeps the change as one short rule.

```
define OEWeakTailNasalLoss [
    {*n} {*ą} -> {*n} || _ .#.,
    {*m} {*ą} -> {*m} || _ .#.
];
```

In prose, the rule reduces final weak-tail endings of the type **-ną* and **-mą* to plain final **-n* and **-m*.

The clearest lexical witness is the pathway to *dōn* 'do'. If the rule is moved too early, before the older reduction that already shapes the **dōną* sequence, the derivation records no output instead of expected OE *dōn* 'do'. No equally sharp later breakpoint appears within the tested sequence. That is why the note remains one-sided and why its earlier relation should be understood as a distant cross-reference only and should not reshape the broader sequence.

This gives the earlier boundary `SC023 < SC061`. No comparably sharp later boundary is available.

The development is best treated as a small late weak-tail adjustment. It remains visible in the sequence because it affects the pathway to *dōn* 'do', but the evidence does not support treating it as the center of a wider historical development.

12 References

Adamczyk, Elżbieta. 2001. 'Old English Reflexes of Sievers' Law'. *Studia Anglica Posnaniensia: An International Review of English Studies* 36: 61–72. http://hdl.handle.net/10593/18335.

Campbell, Alistair. 1959. *Old English Grammar*. Clarendon Press.

Fulk, R. D. 2018. *A Comparative Grammar of the Early Germanic Languages*. Vol. 3. Studies in Germanic Linguistics. John Benjamins.

Hogg, Richard M. 1992. *A Grammar of Old English. Volume 1: Phonology*. Blackwell.

Luick, Karl. 1914--1940. *Historische Grammatik Der Englischen Sprache*. Tauchnitz.

Ringe, Don, and Ann Taylor. 2014. *The Development of Old English*. Vol. 2. A Linguistic History of English. Oxford University Press.